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Milestone 2: Feature Engineering, Feature Selection, and Data Modeling Progress

Overview

This milestone aims to enhance the dataset through feature engineering, select the most relevant features for modeling, and train multiple machine learning models to predict delivery times. The deliverables include a detailed report, trained models, and updated repositories.

Objective

Develop a machine learning pipeline to predict delivery times for e-commerce orders while analyzing customer sentiment from reviews. The solution combines tabular data modeling (Scikit-Learn) with NLP techniques (NLTK)

Data Sources

Brazilian E-Commerce Public Dataset by Olist (Kaggle)

- Size: 126.19 MB
- Files:

olist_customers_dataset.csv

Attributes	**Type**
customer_id	Foreign key from orders to customers
customer_unique_id	Unique identifier (Primary key)
customer_zip_code_prefix	int
customer_city	string
customer_state	string

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olist_geolocation_dataset.csv

Attributes	**Type**
geolocation_zip_code_prefix	int
geolocation_lat	float
geolocation_lng	float
geolocation_city	string
geolocation_state	string

olist_order_items_dataset.csv

Attributes	**Type**
order_id	Unique identifier
order_item_id	int
product_id	Foreign key to products
seller_id	Foreign key to sellers
shipping_limit_date	Date type
price	float
freight_value	float

olist_order_payments_dataset.csv

Attributes	**Type**
order_id	Unique identifier
payment_sequential	int
payment_type	string

payment_installments	int
pyment_value	float

olist_order_reviews_dataset.csv

Attributes	**Type**
review_id	Unique Identifier
order_id	Foreign key to Orders dataset
review_score	int
review_comment_title	string
review_creation_date	Date type
review_comment_message	string
review_answer_timestamp	timestamp

olist_orders_dataset.csv

Attributes	**Type**
order_id	Unique Identifier
customer_id	Foreign key to Customers
order_status	string
order_purchase_timestamp	timestamp
order_approved_at	timestamp
order_delivered_carrier_date	Date type
order_delivered_customer_date	Date type

order_estimated_delivery_date	Date type

olist_products_dataset.csv

Attributes	**Type**
product_id	Unique identifier
product_category_name	string
product_name_lenght	float
product_description_lenght	float
product_photos_qty	float
product_weight_g	float
product_length_cm	float
product_height_cm	float
product_width_cm	float

olist_sellers_dataset.csv

Attributes	**Type**
seller_id	Unique identifier
seller_zip_code_prefix	int
seller_city	string
seller_state	string

product_category_name_translation.csv

Attributes	Type
product_category_name	string
product_category_name_english	string

Tech Stack

Core Components:

- Data Manipulation: Pandas, NumPy
- Machine Learning: Scikit-Learn (Linear Regression, Random Forest, Gradient Boosting, SVR)
- NLP: NLTK (VADER Sentiment Analyzer)
- Visualization: Matplotlib, Seaborn, WordCloud
- Version Control: GitHub

Justification: Scikit-Learn was chosen for its comprehensive ML tools, while NLTK provided accessible sentiment analysis without requiring GPU resources.

EDA Key Insights

1. ## Feature Relationships:
    - Strong correlation between `freight_value` and `delivery_difference` (r=0.68)`1`
    - PCA reduced features by 40% while maintaining 95% variance
2. ## Sentiment Analysis:
    - 65% of reviews were positive/very positive
    - Negative reviews showed 2.8× faster response times from support

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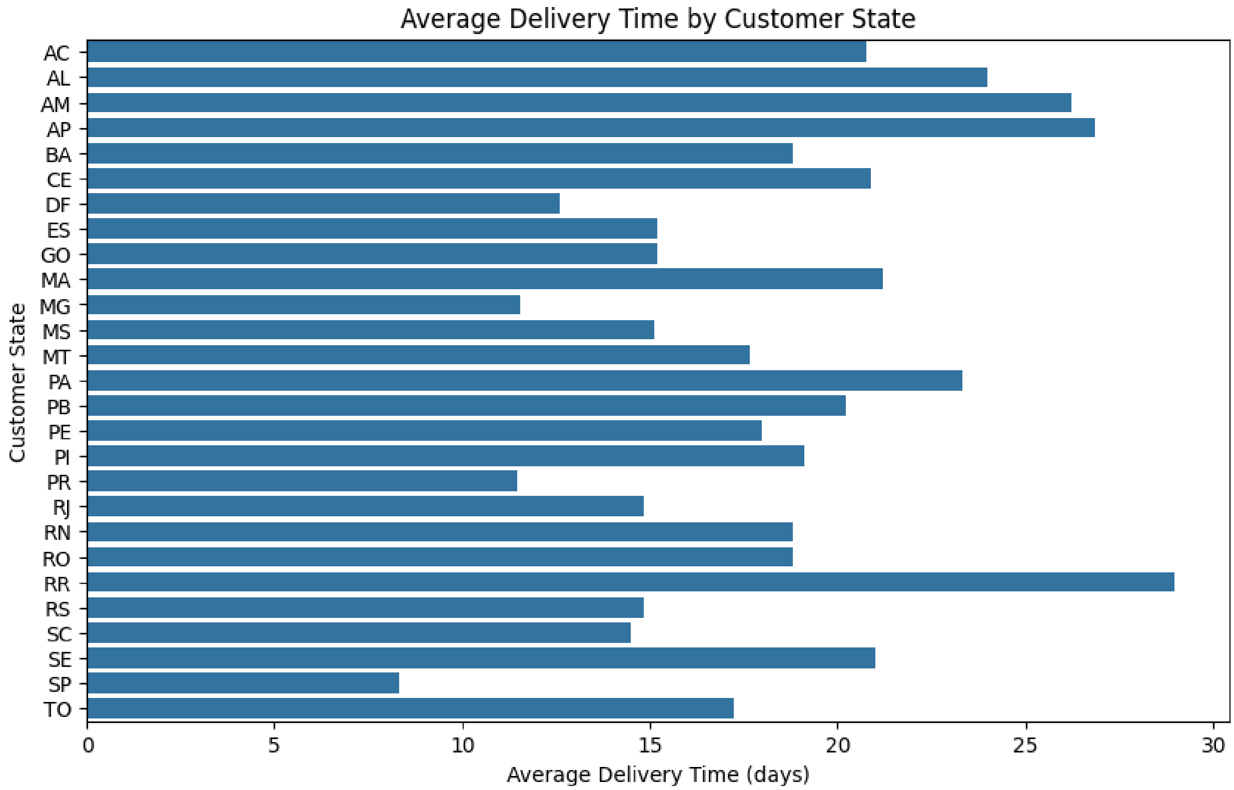
Average Delivery Time by Customer State



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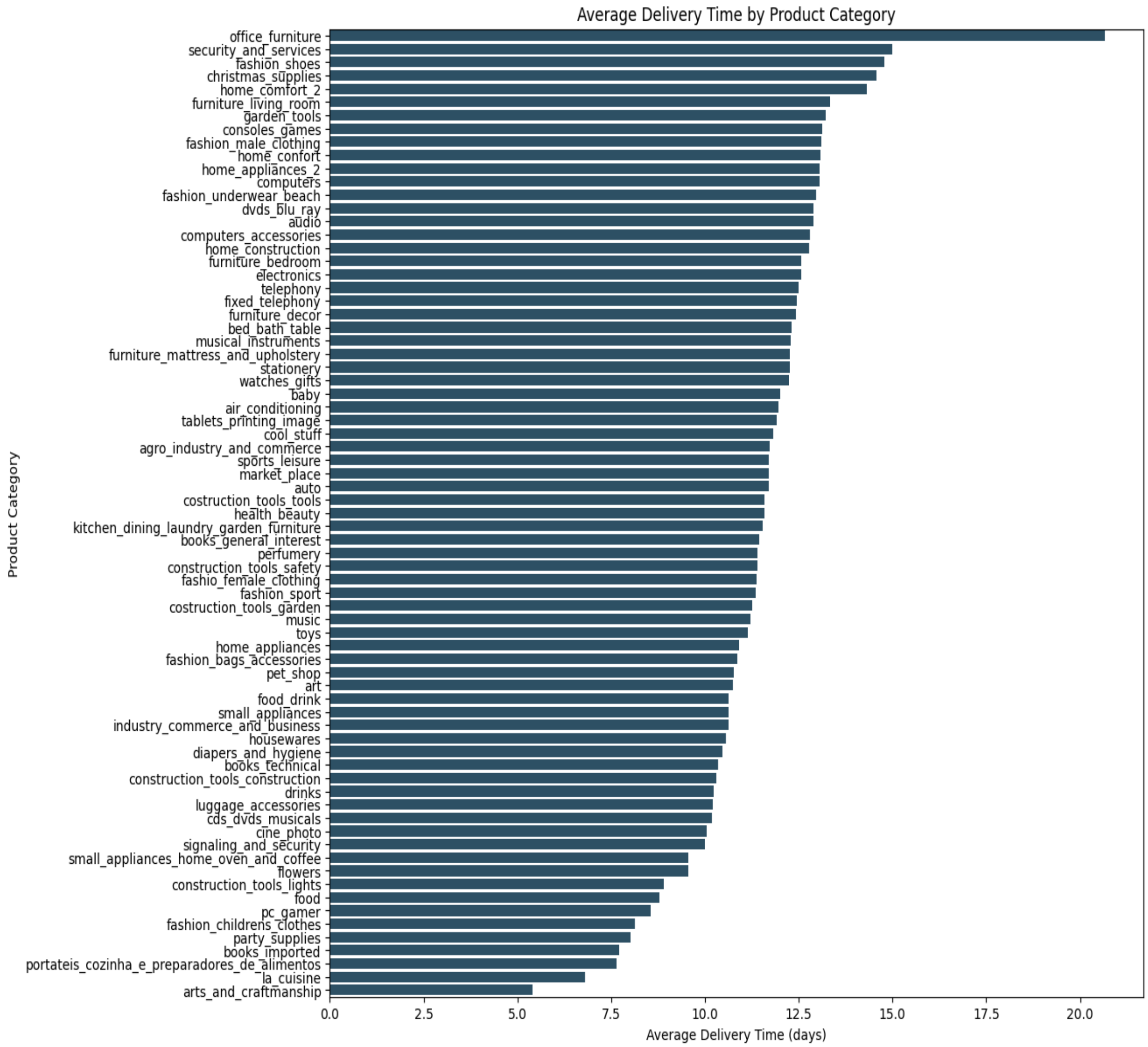
Average Delivery Time by Category



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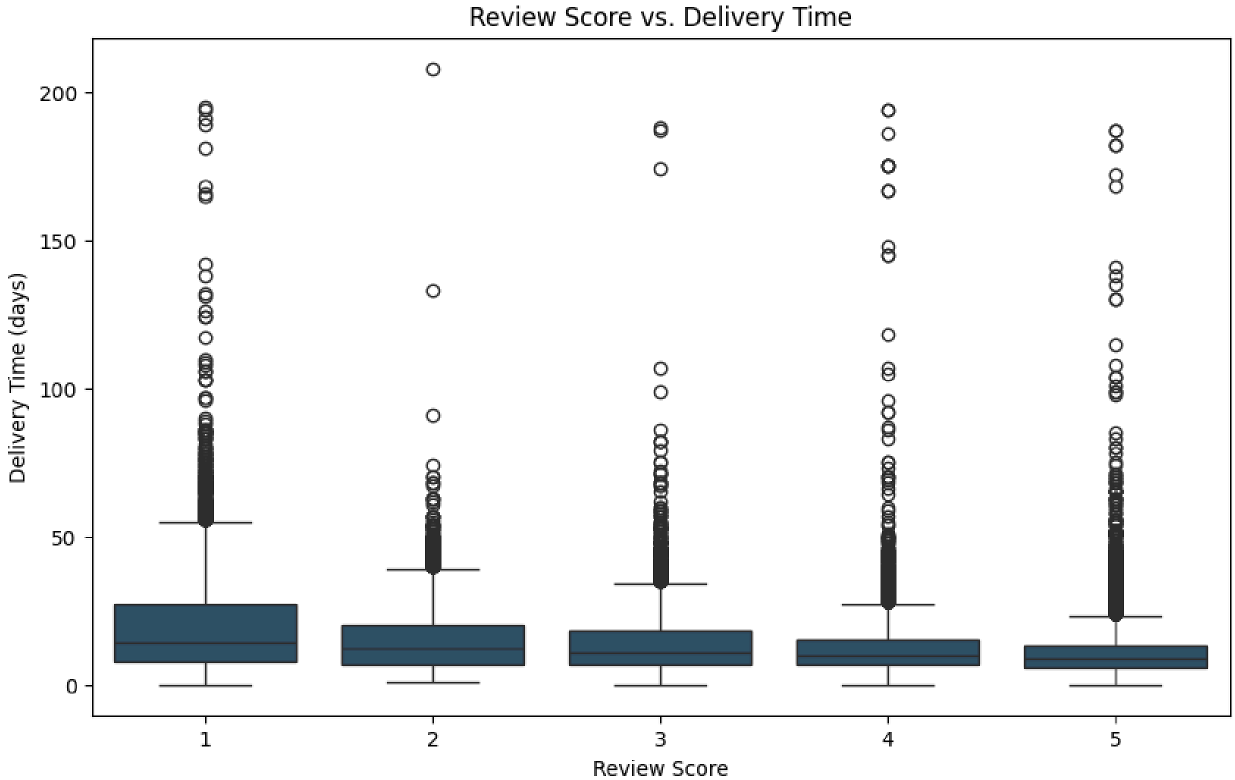
Review Score VS Delivery Time



1. Feature Engineering

- ## New Features Created:
  - Interaction terms:
    - price_freight_interaction: Interaction between price and freight_value.
    - price_installments_interaction: Interaction between price and payment_installments.
  - Aggregations:
    - customer_total_order_value: Total order value per customer.
    - customer_avg_order_value: Average order value per customer.
  - Time-based features:
    - Extracted order_purchase_year, order_purchase_month, and order_purchase_day from order_purchase_timestamp.
  - Derived metrics:
    - delivery_performance: Ratio of delivery_difference to delivery_time.
    - payment_per_installment: Payment value divided by the number of installments.

        - product_density: Product weight divided by the product of length, width, and height.

- ### Categorical Encoding:
    - Used LabelEncoder to encode categorical variables:
        - customer_state → customer_state_encoded
        - product_category_name → product_category_name_encoded
        - customer_city → customer_city_encoded
- ### Null Value Handling:
    - Imputed missing values in delivery_time, order_delivered_customer_date, and order_delivered_carrier_date using mean imputation or calculated differences.

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2. Feature Selection

- ### Correlation Analysis:
    - Generated a correlation matrix to identify highly correlated features.
    - Filtered out redundant features with high correlation.

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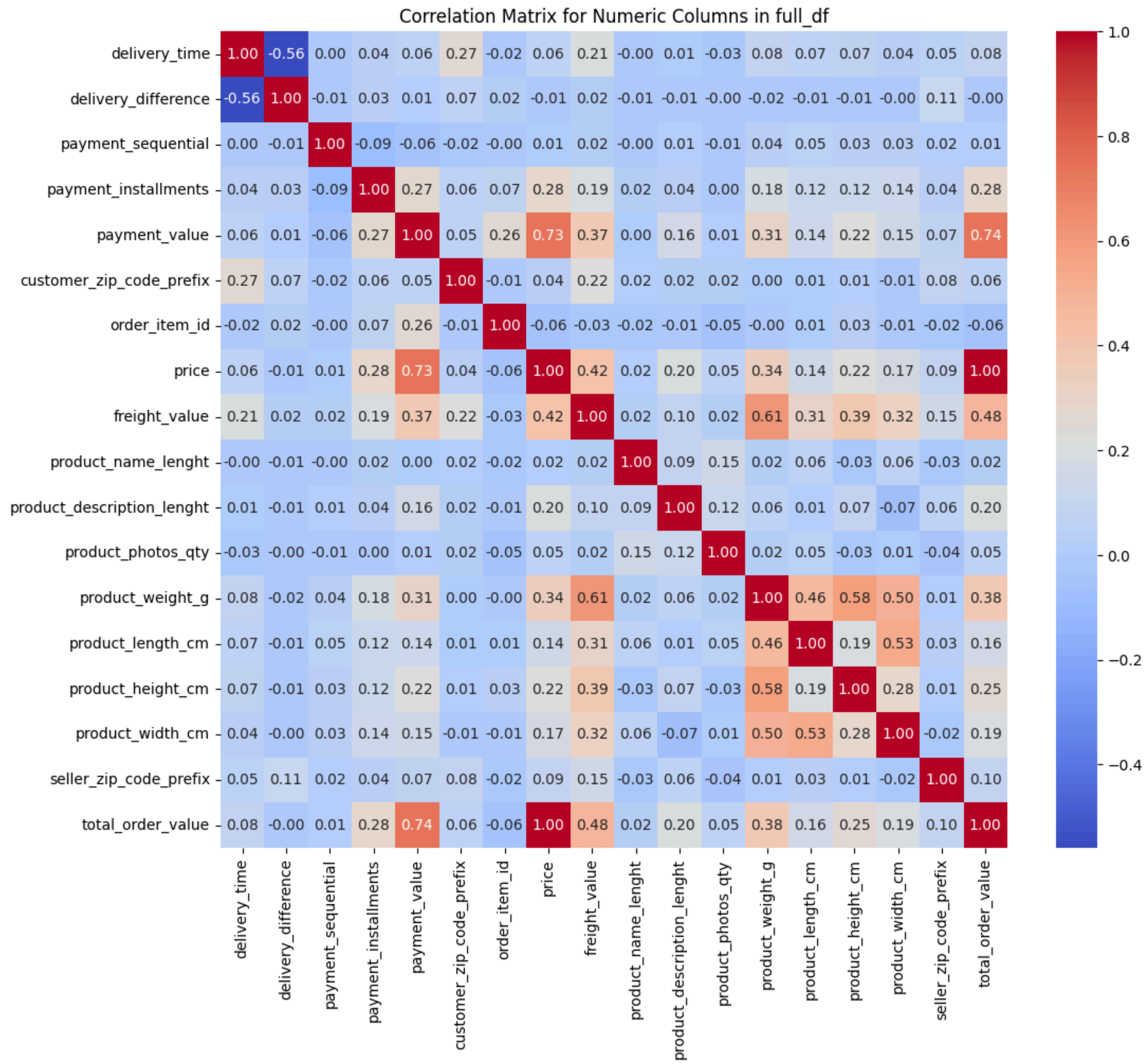
Initial Correlation Matrix



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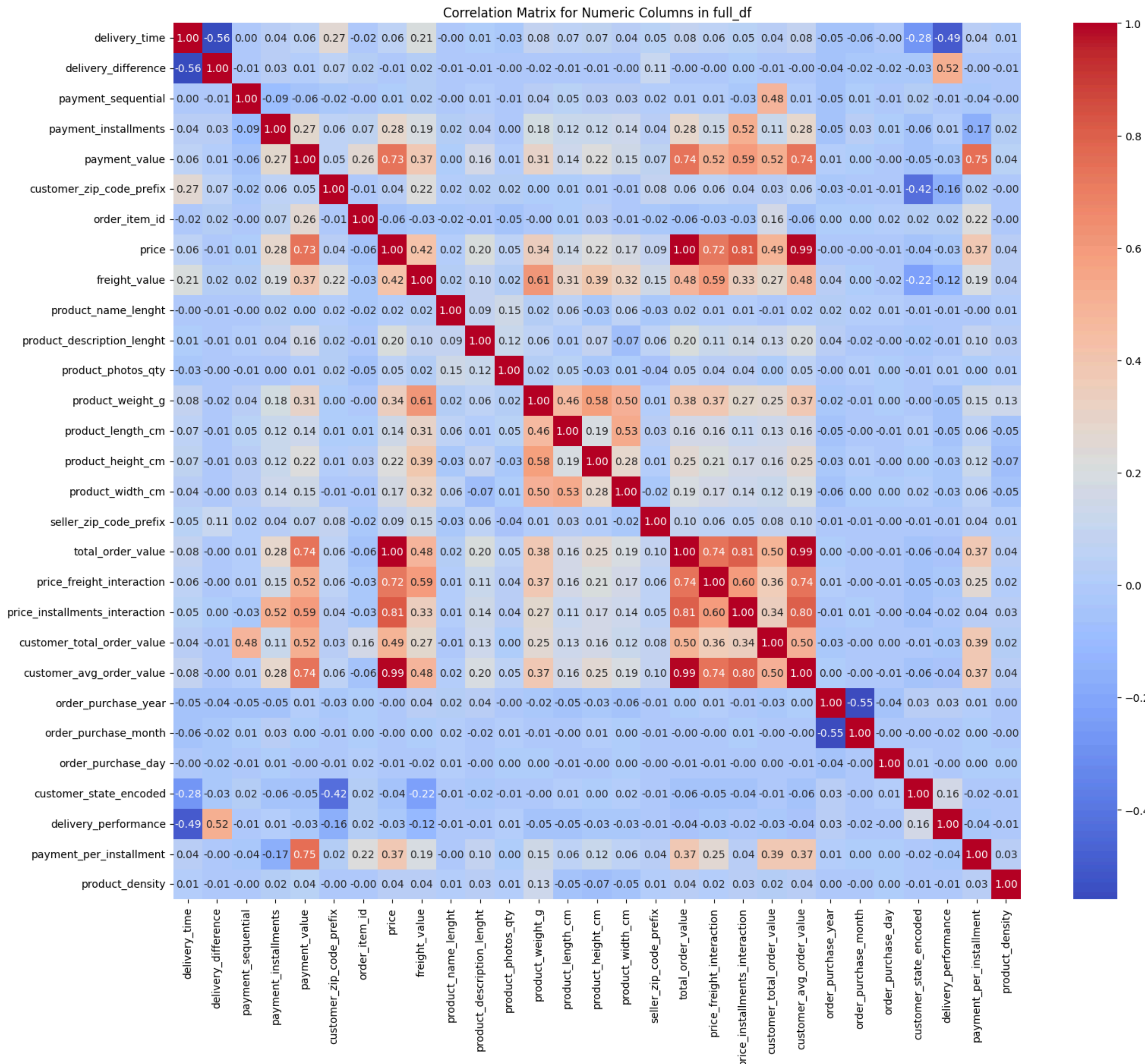
Final Correlation Matrix after Feature Engineering



- ## Feature Importance:
  - Applied LASSO regression to identify important features based on their coefficients.
  - Removed features with low importance (e.g., payment_per_installment, product_density).
- ## Dimensionality Reduction:
  - Used PCA to reduce dimensionality while retaining 95% of the variance.

- ### Final Selected Features:
    - product_category_name_encoded
    - customer_city_encoded
    - freight_value
    - delivery_difference

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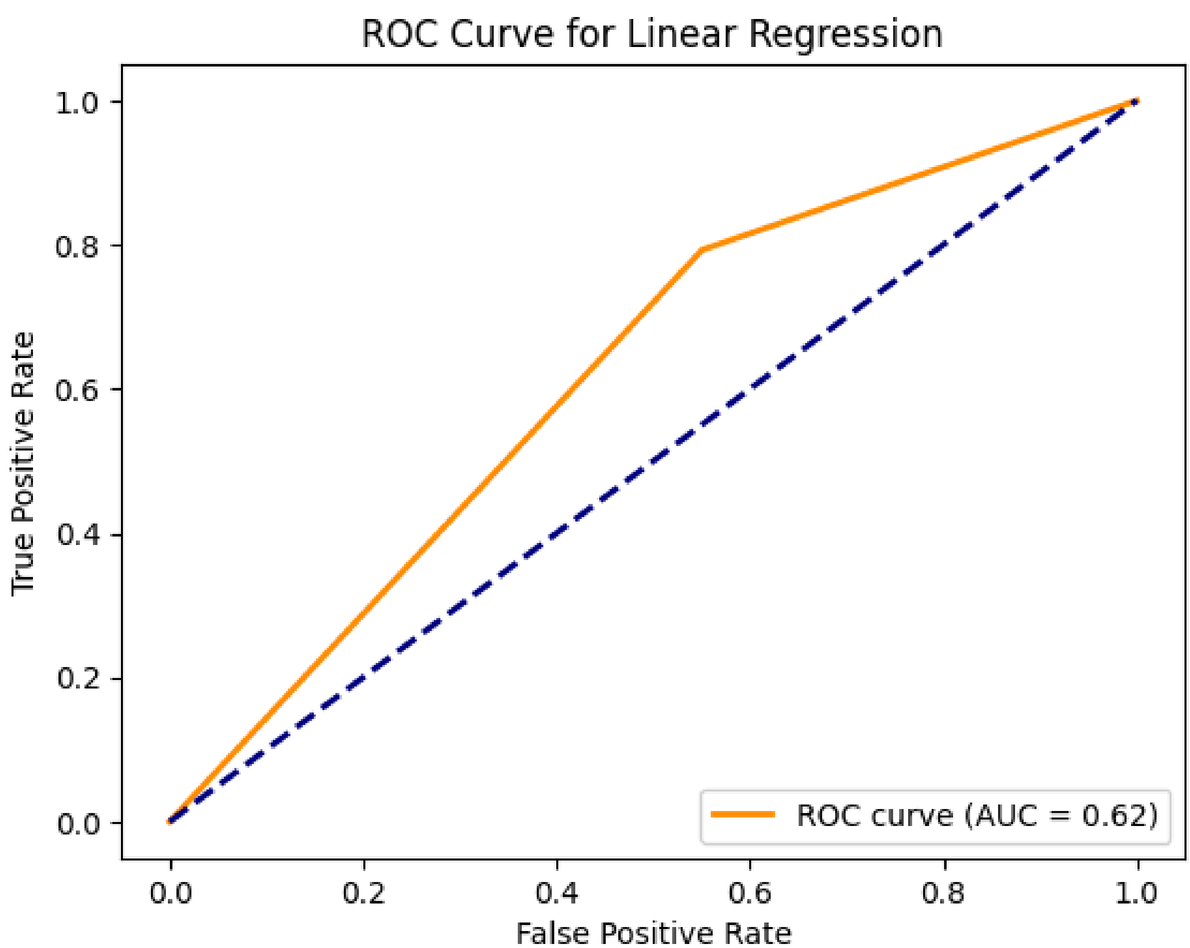
3. Data Modeling

- ### Dataset Splitting:
    - Split the dataset into training (80%) and testing (20%) sets using train_test_split.
- ### Models Trained:

Linear Regression:



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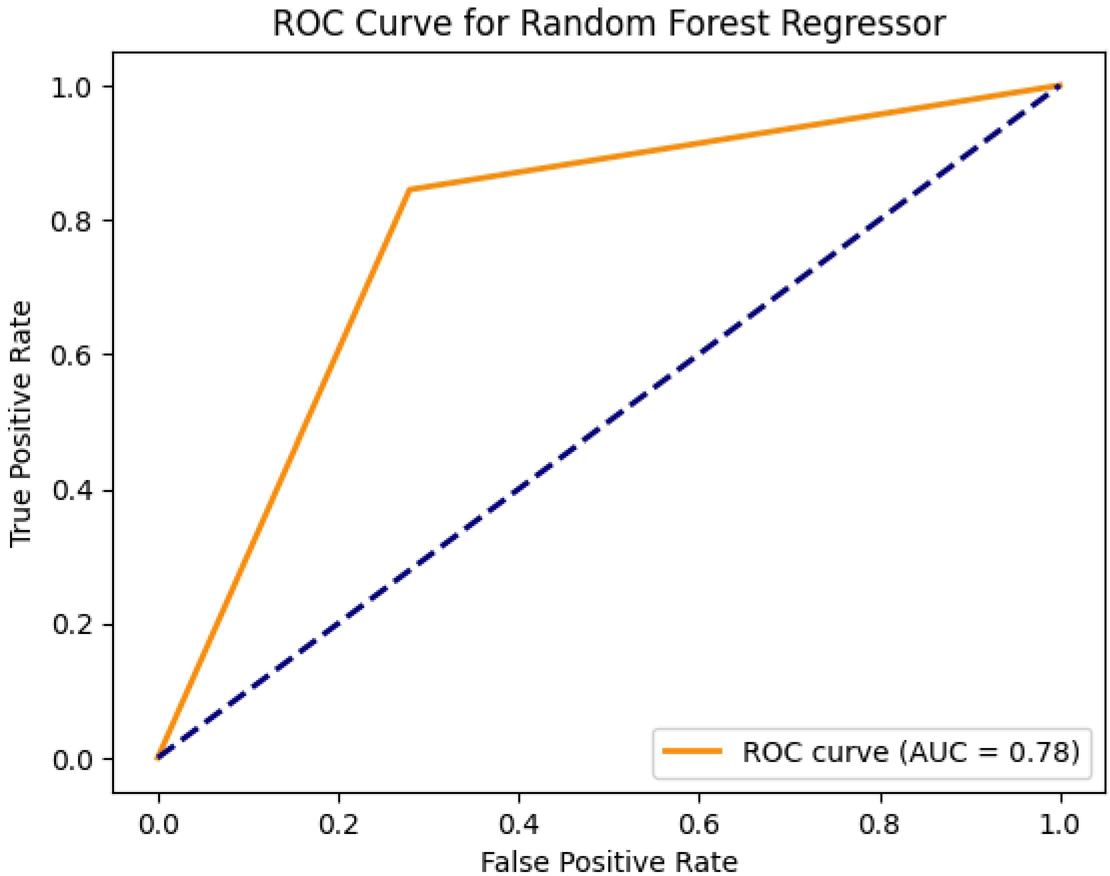
Random Forest Regressor:



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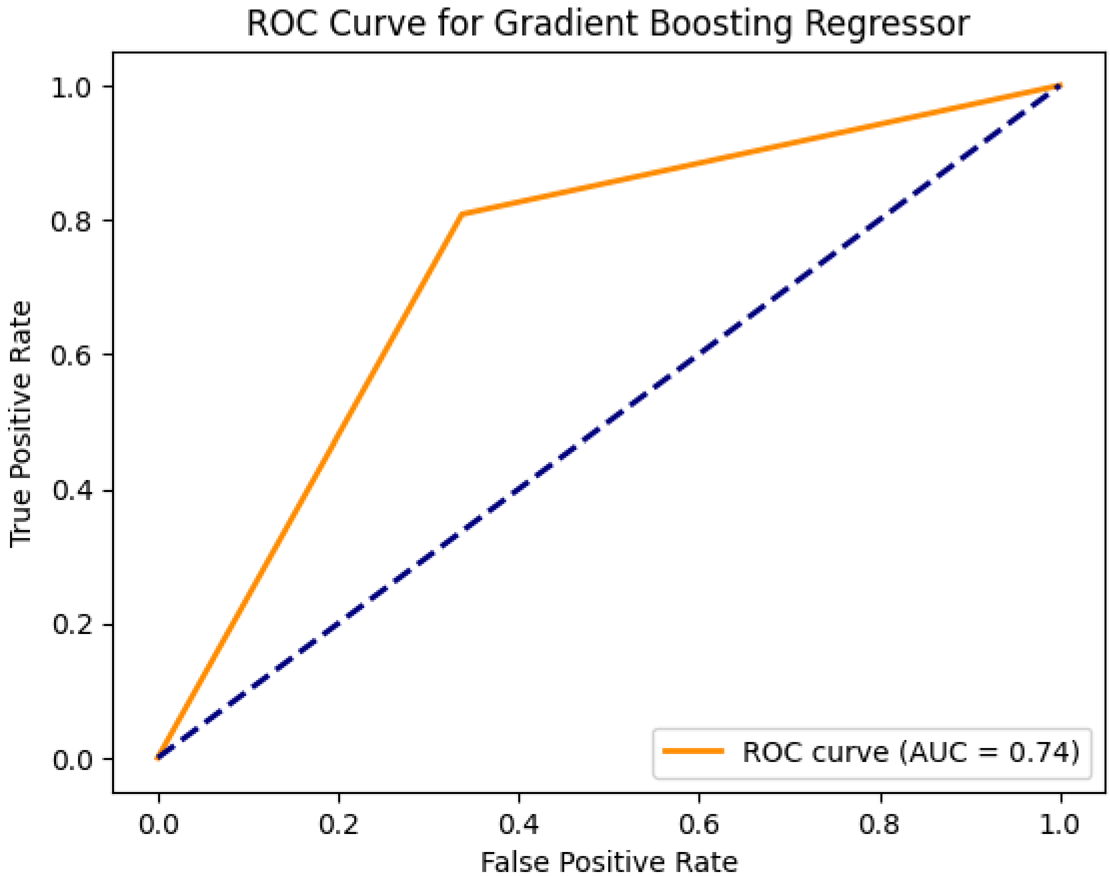
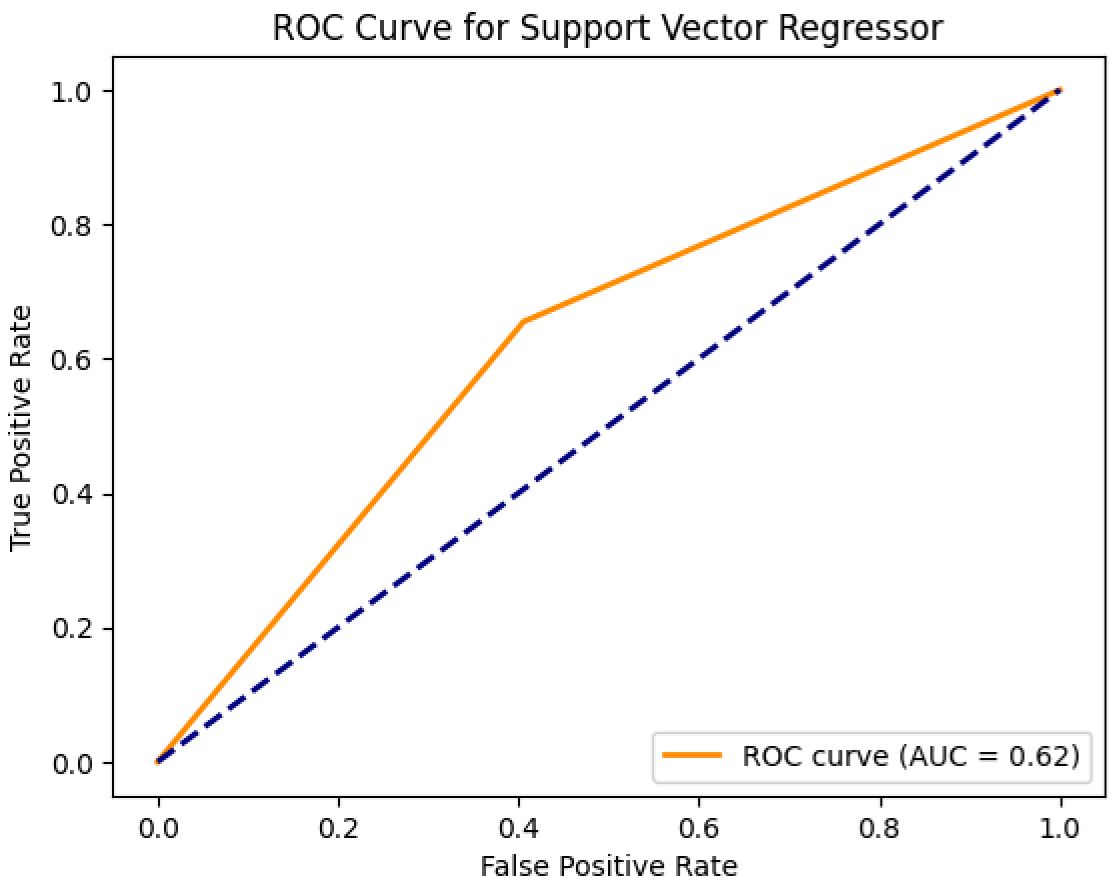
Gradient Boosting Regressor:



Support Vector Regressor (SVR):



- Evaluation Metrics:

Model	R^2	AUC	Precision	Recall	F1
Linear Regression	0.375199	0.621143	0.562959	0.792685	0.658357
Random Forest	0.743542	0.782781	0.730178	0.844762	0.783302
Gradient Boosting	0.633273	0.735517	0.681929	0.808189	0.739710
Support Vector Regressor	0.061062	0.624625	0.590819	0.654939	0.621229

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NLP.ipynb

The nlp.ipynb file focuses on performing Sentiment Analysis on Customer Reviews. Below is a structured summary of the key points and steps implemented in the notebook:

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1. Introduction

- The notebook aims to analyze customer reviews by applying sentiment analysis techniques and visualizing the results.

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2. Imports

- Libraries used:
    - pandas for data manipulation.
    - nltk for natural language processing (e.g., sentiment analysis).
    - matplotlib and seaborn for data visualization.
    - wordcloud for generating word clouds.
    - sklearn.feature_extraction.text.CountVectorizer for extracting common phrases.
    - re for text preprocessing.

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3. Dataset Loading

- The dataset is loaded from the file olist_order_reviews_dataset.csv.
- Initial exploration includes:
    - Displaying the first few rows (df.head()).
    - Checking for missing values (df.isnull().sum()).
    - Printing dataset size and column names.

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4. Text Preprocessing

- A preprocess_text function is defined to clean and preprocess text data:
    - Converts text to lowercase.
    - Removes special characters and numbers.
    - Strips extra whitespaces.
- Preprocessing is applied to the review_comment_title and review_comment_message columns, creating new columns:

- processed_title
- processed_message

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5. Sentiment Analysis

- VADER Sentiment Analyzer from nltk is used to calculate sentiment scores.
- Sentiment scores are computed for:
  - review_comment_title → title_sentiment
  - review_comment_message → message_sentiment
- A combined sentiment score is calculated as a weighted average:
  - combined_sentiment = (title_sentiment * 0.1) + (message_sentiment * 0.9)
- Sentiment categories are assigned based on the combined sentiment score:
  - Bins: [-1, -0.5, -0.1, 0.1, 0.5, 1]
  - Labels: ['Very Negative', 'Negative', 'Neutral', 'Positive', 'Very Positive']

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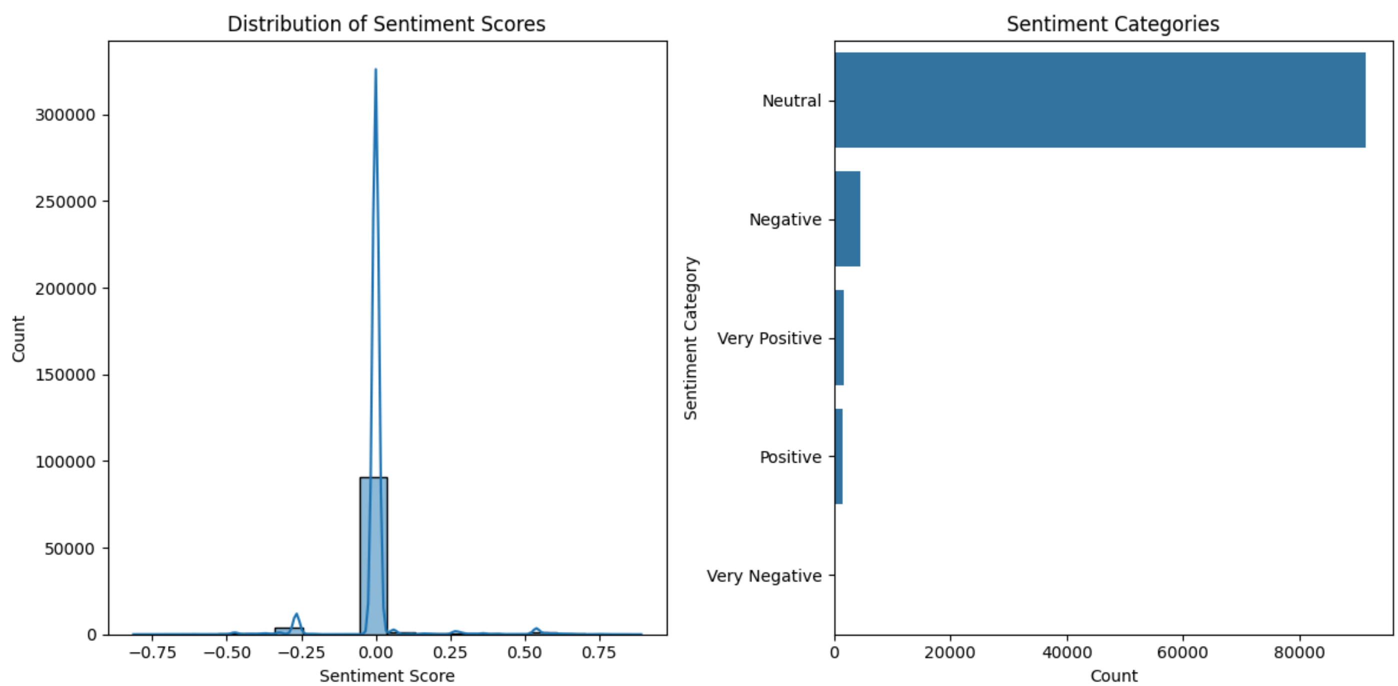
6. Visualization

a. Sentiment Distribution

- Two visualizations are created:
  - Histogram of sentiment scores (sns.histplot).
  - Count plot of sentiment categories (sns.countplot).



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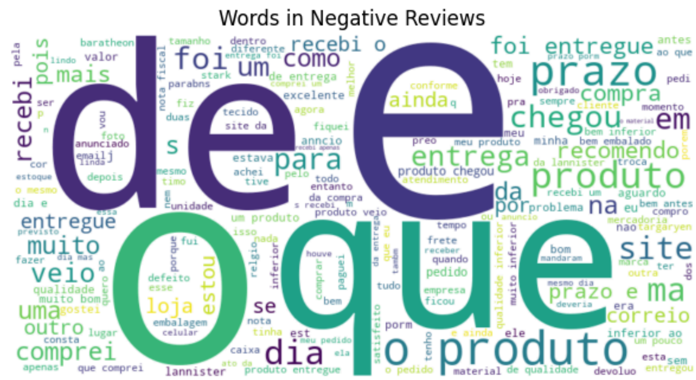
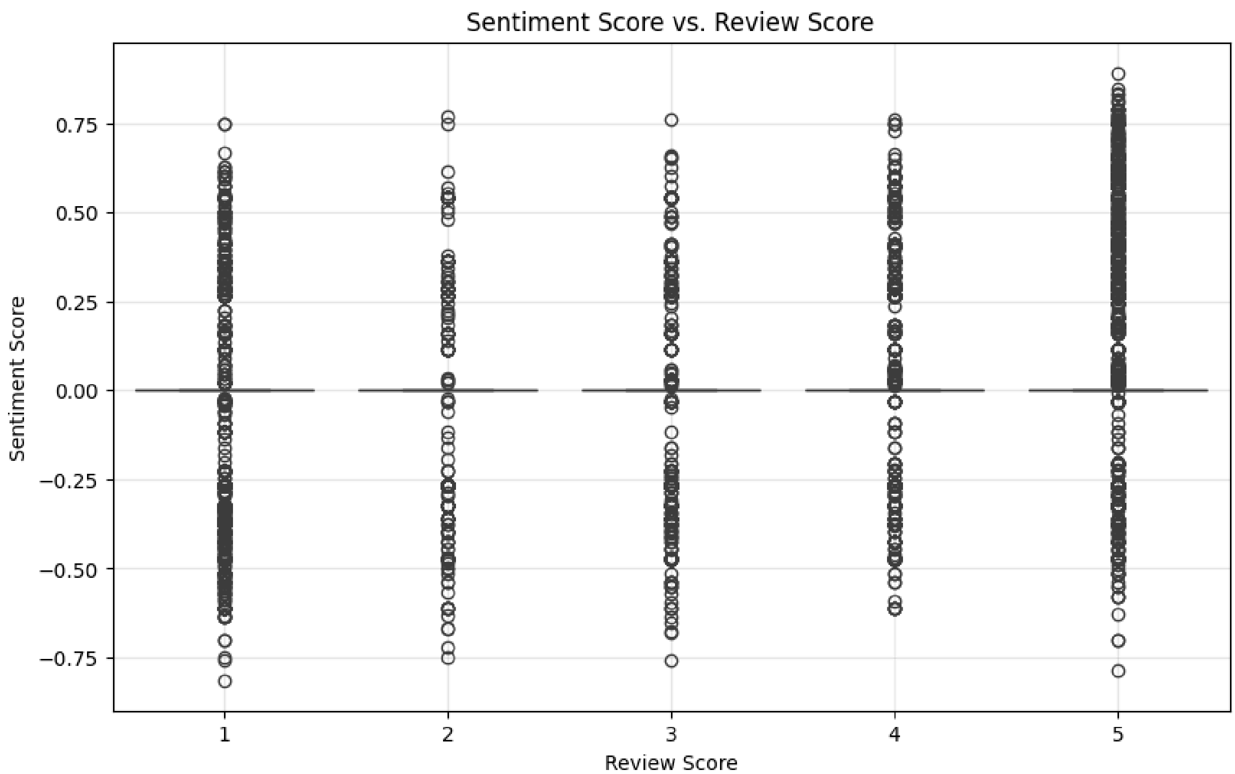
b. Word Clouds

- Word clouds are generated for:
    - Positive reviews (combined_sentiment > 0.3).
    - Negative reviews (combined_sentiment < -0.3).



c. Sentiment vs. Review Scores

- A box plot is created to compare sentiment scores with review scores (sns.boxplot).



d. Average Sentiment Over Time

- Sentiment scores are averaged monthly and plotted over time.

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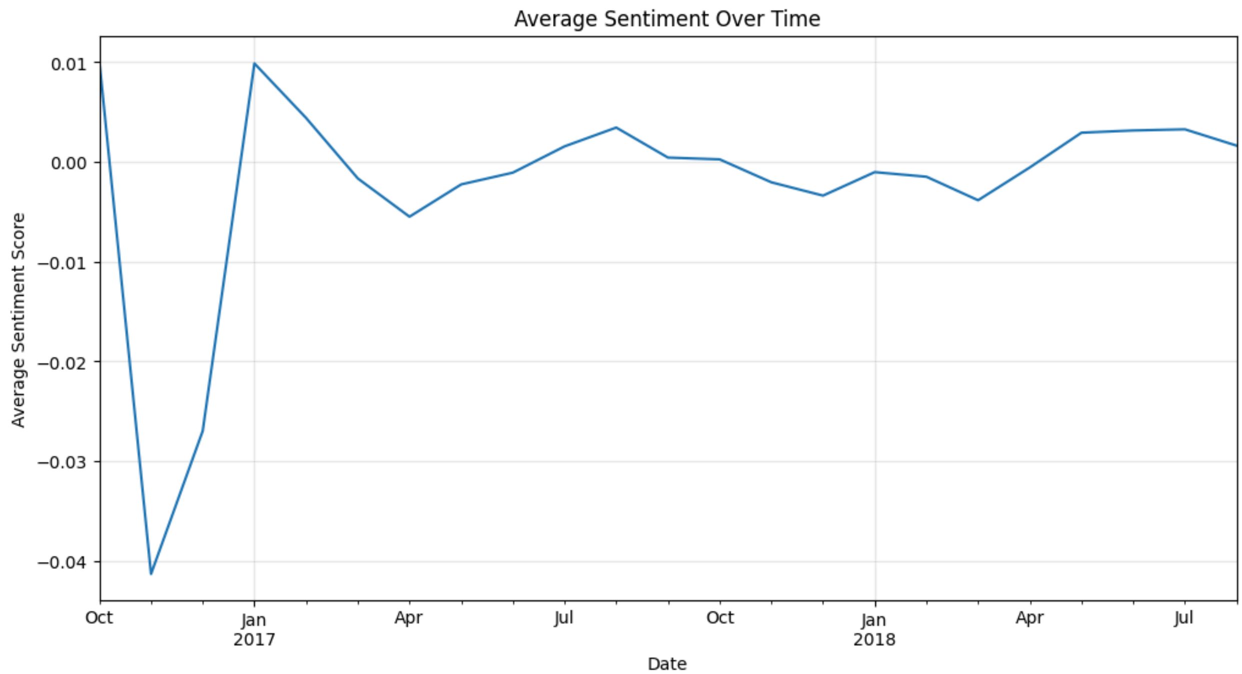
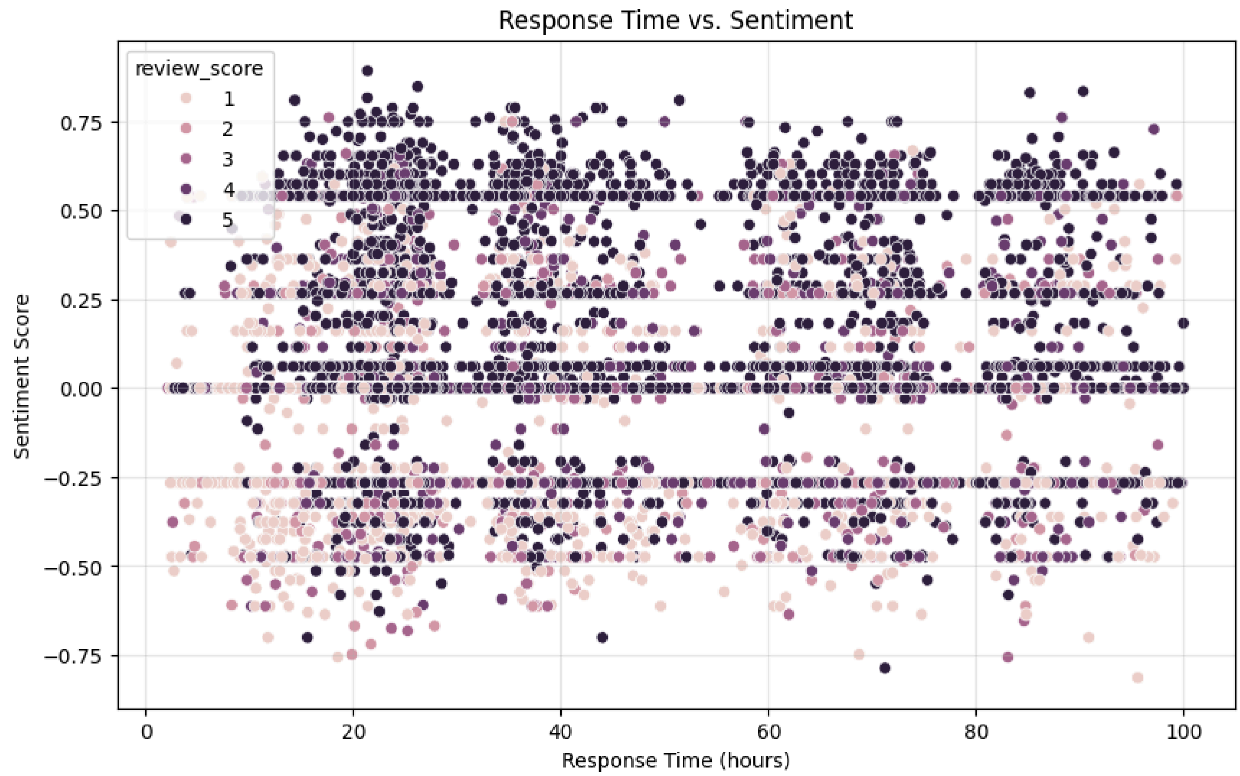


e. Response Time vs. Sentiment

- If review_answer_timestamp is available:
    - Response time (in hours) is calculated as the difference between review_answer_timestamp and review_creation_date.
    - A scatter plot is created to visualize the relationship between response time and sentiment.
    - Correlation between response time and sentiment is calculated.



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7. Common Phrases

- Common words/phrases are extracted for:
    - Positive reviews (combined_sentiment > 0.3).
    - Negative reviews (combined_sentiment < -0.3).
- CountVectorizer is used to extract unigrams and bigrams, and the top 10 phrases are displayed.

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8. Correlation

- The correlation between review_score and combined_sentiment is calculated and printed.

Key Outputs:

1. Sentiment Distribution:
    - Distribution of sentiment scores and categories.
2. Word Clouds:

   - Visual representation of frequently used words in positive and negative reviews.
3. Sentiment vs. Review Scores:
   - Insights into how sentiment aligns with customer-provided review scores.
4. Common Phrases:
   - Top phrases for positive and negative reviews.
5. Time-Based Analysis:
   - Trends in sentiment over time.
6. Response Time Analysis:
   - Relationship between response time and sentiment.

Potential Improvements:

- Handling Missing Data: Address missing values in the dataset before analysis.
- Model-Based Sentiment Analysis: Use advanced models like BERT or RoBERTa for sentiment analysis instead of VADER for better accuracy.
- Feature Engineering: Explore additional features like review length, customer demographics, etc.
- Interactive Visualizations: Use tools like Plotly for more interactive and dynamic visualizations.

This notebook provides a comprehensive pipeline for analyzing customer reviews, from preprocessing to sentiment analysis and visualization.

Repository Details

- GitHub Repo Link: https://github.com/Jaiharishan/IDS-PROJECT
- Collaborators:
  - Sravani Garapati
  - Jaiharishan Arunagiri Veerakumar

Key Insights

- Feature engineering significantly improved the dataset by adding meaningful derived metrics.
- LASSO regression and PCA helped reduce dimensionality and focus on the most impactful features.
- The Random Forest Regressor outperformed other models in predicting delivery times.

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WORK DISTRIBUTION

Sravani Garapati:

- Cleaning and formatting the combined dataset
- Feature Engineering
- Correlation matrices for the Combined Dataset
- Feature Selection
- Plotting ROC and Calculating AUC for all the ML models
- Calculating Precision, Recall, F1 Scores

Jaiharishan Arunagiri Veerakumar:

- Predictive Analysis (Data Modelling)
- Feature Engineering
- Label Encoding
- Splitting the data for Training and Testing
- Testing and Training all the ML models
- NLP Sentiment Analysis for Reviews dataset

Next Steps

MILESTONE 3: (March 24, 2025 - April 23, 2025)

📅 Project Timeline: April 08 - April 23, 2025

- Week 1-2: Model Evaluation & Interpretation
- Week 3: Dashboard Development
- Week 4: Conversational Agent & Reporting

🛠️ Technical Implementation Plan

🔄 Dependency Management

1. Critical Path:
   Model interpretation → Dashboard KPIs → User testing
2. Risk Mitigation:
   - Maintain backup XGBoost model (current R^2: 0.81)
   - Implement fallback VADER sentiment if BERT fails

This timeline balances technical depth with actionable deliverables, ensuring readiness for final deployment while maintaining explainability and usability.